DANIEL SANTANDER URRUTIA

Environmental Geophysicist & Catchment Modeler • Dual M.Sc. Candidate

EXECUTIVE RESUME (1-PAGE)

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Executive Profile: Quantitative environmental geophysicist with dual European Master of Science degrees (CAU Kiel, Germany & AMU Poznań, Poland) and a Geophysics degree (awarded with Highest Distinction) from Universidad de Chile. Combines rigorous subsurface hydrogeophysical modeling (ERT, VES, piezometry) and atmospheric GHG flux quantification (Eddy Covariance towers, dynamic flux chambers) with full-stack scientific software engineering (Python, R, QGIS, PostGIS, Docker, React). Proven track record leading nationwide transmedia campaigns (+700k reach) and testifying before Chile's National Congress on wildfire disaster propagation and watershed resilience.

OFFICIAL HIGHER EDUCATION & UNIVERSITY DEGREES

3 ACCREDITED DEGREES

M.Sc. in Environmental Management — Christian-Albrechts-Universität zu Kiel (CAU Kiel) Kiel, Germany | 2023 – 2026

Faculty of Agricultural & Nutritional Sciences. Research on peatland rewetting biophysics (Wallener Au), catchment water balance, soil precompression stress (σ_p = 62.4 kPa), and Eddy Covariance GHG flux dynamics (CO_2, CH_4). Developed the **Wetland Monitor** 3D digital twin.

M.Sc. in Environmental Protection — Adam Mickiewicz University (AMU Poznań) Poznań, Poland | 2023 – 2026

Faculty of Geographical & Geological Sciences. Specialization in post-glacial lake carbon dynamics, microscopic freshwater bioindicators, aquatic ecosystem degradation, and compliance under the EU Water Framework Directive.

Geophysicist & B.Sc. in Geophysics — Universidad de Chile Santiago, Chile | 2015 – 2021

Faculty of Physical & Mathematical Sciences (FCFM). **Awarded with Highest Distinction**. Applied geophysics, solid Earth physics, fluid mechanics, seismology, and groundwater modeling. Degree thesis on soil electrical resistivity (VES/SEV) and water table drawdown in the Maule Basin under agroforestry and climate change.

KEY PROFESSIONAL & RESEARCH EXPERIENCE

FIELD, MODELING & LEADERSHIP

Developer & Author — Wetland Monitor (3D web application) 2026

- Designed and built an interactive 3D web application presenting the quantitative results of the Wallener Au peatland rewetting study (CH_4 sink-to-source transition, Eddy Covariance, paludiculture, soil compaction), with a day/seasonal simulator.

Lead Technical Author & Policy Analyst — Environmental Paper Network (EPN) International | 2023

- Authored international case study *"Spotlight on Burning Biomass"* (Parral, Chile), exposing carbon accounting loopholes and auditing European renewable energy subsidies under EU RED III.

Parliamentary Expert Spokesperson — National Congress of Chile (Chamber of Deputies) Valparaíso, Chile

- Presented empirical hydrogeophysical data before the Special Commission on Disasters and Wildfires, demonstrating the causal link between pine/eucalyptus plantations, water drawdown, and fire speed.

Communications Director & Campaign Coordinator — Apruebo Rural & Tactical Media Chile | 2020 – 2023

- Directed a national public interest campaign reaching +700,000 citizens across digital networks and community assemblies, translating complex legal, ecological, and water policy proposals into accessible visual media.

CORE TECHNICAL MATRIX & OPERATIONAL CAPABILITIES

VERIFIED STACK

GEOSPATIAL & DATA SCIENCE

Languages: Python (GeoPandas, NumPy, SciPy, Rasterio, GDAL), R (sf, ggplot2), SQL (PostgreSQL/PostGIS), LaTeX.
Engines & GIS: QGIS, ArcGIS Pro, Google Earth Engine, Docker, Git/GitHub, Delft3D coastal surge modeling.

FIELD GEAR & WEB SYSTEMS

Instrumentation: Syscal Pro (ERT), LI-COR (Eddy Covariance CO_2/CH_4), Dynamic soil flux chambers, CTD divers, Edelman coring.
Web & Full-Stack: TypeScript, React, Next.js, Tailwind CSS, FastAPI, Three.js, REST APIs.

Languages: Spanish (Native, parliamentary speaking) • English (C1 Fluent, international keynote & academic writing) • German (B1-B2 Working).

Key Summits: WSL Davos (2025), DEAL Copenhagen (2025), IOER Dresden (2024).